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Feasibility of Real-Time Automated Vocal Fold Motion Tracking for In-Office Laryngoscopy

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Abstract

Objectives: Major advancements have been made in applying artificial intelligence and computer vision to analyze videolaryngoscopy data. These models are limited to post hoc analysis and are aimed at research settings. In this work, we assess the feasibility of a real-time solution for automated vocal fold tracking during in-office laryngoscopy.

Methods: We trained a keypoint detection model that tracks 39 individual keypoints of the larynx, optimized for real-time deployment, on 1254 frames extracted from 57 laryngoscopy videos. Images were annotated with keypoints along the true and false vocal folds, and supraglottic structures. After training, the model was evaluated on a validation dataset of 140 test images, and second independent test dataset of 50 images. Performance was assessed by comparing manual keypoint annotations to predicted annotations and computing mean keypoint accuracy (mKA), and by calculating temporal consistency (mTC) from a calibration video clip. The live feed of a flexible laryngoscope, mounted to an endoscopic tower, was passed via a video capture card into a laptop running the motion tracking model in real-time.

Results: The model demonstrated robust detection accuracy with an mKA of 85% and 75% with validation and test datasets, and promising temporal consistence mTC of 5.85 average pixel movement. During testing, the model ran at 30 frames per second, with minimal appreciable latency while processing the laryngoscope's live feed, ensuring the laryngologist's work was not affected.

Conclusion: The model accurately tracks laryngeal structures with strong detection, localization, and consistency, is compatible with affordable hardware, and enables real-time metric development.

Level of Evidence: 4.

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Keywords

artificial intelligence; videolaryngoscopy; vocal fold tracking

1 | Introduction

Laryngoscopy is commonly used to diagnose voice disorders [1], swallowing difficulties [2], breathing issues [3] through the detection of anatomic or functional abnormalities [4, 5]. Traditionally, videolaryngoscopy examinations are performed by medical staff, who inspect the video feed in real time to assess laryngeal function. While recordings can be made for documentation, detailed post hoc or computational analysis is uncommon in routine clinical practice and typically reserved for research settings. As a result, the assessment largely relies on the clinician's experience and subjective interpretation [6], with limited support from automated or quantitative analysis tools in standard practice. This gap in analysis highlights the need for more objective and scalable approaches to support clinical decision making.

Recent advancements in artificial intelligence (AI) and computer vision (CV) have demonstrated promising potential in automating medical image analysis in this field [7–11]. These include keypoint tracking for glottal opening angle estimation by our group [12–14], vocal fold image segmentation [15, 16], pose estimation [17], and glottal closure estimation [18]. AI models have also achieved high accuracy in detecting anatomical structures and tracking motion in dynamic clinical videos where clear pathological outcomes, such as laryngeal cancer, present additional challenges [19]. However, most existing AI applications in videolaryngoscopy are designed for specific use-cases and offline analysis, which limits their utility for real-time clinical decision-making during live examinations.

Integrating AI, specifically CV, into videolaryngoscopy has the potential to transform voice diagnostics by offering immediate feedback during procedures, improving accuracy, and reducing subjectivity. However, bringing such capabilities into real-time clinical use poses distinct technical and practical challenges. Technically, the CV model must be both accurate and stable enough to provide clinically meaningful predictions, while also being lightweight enough [20] to run on standard computer hardware commonly available in clinical labs, avoiding the need for costly computational infrastructure. Practically, the real-time predictions must be delivered with minimal delay, so that the examining physician can interpret and act on the information seamlessly during the procedure. Despite growing interest in AI technologies, the feasibility of real-time AI-based vocal fold motion tracking during laryngoscopy remains underexplored.

In this study, we present a proof-of-concept implementation of a real-time CV model for vocal fold motion tracking in in-office laryngoscopy. We outline the development, training, and deployment of a lightweight keypoint detection model that tracks an expanded set of individual laryngeal keypoints with high accuracy and temporal consistency while operating at 30 frames per second (fps) on a portable device, such as a laptop. This work was motivated by the need to support future clinical applications beyond glottal opening, such as assessing supraglottic activity, false vocal cord involvement, and midline symmetry. Our aim was to evaluate whether such a tracking model could be feasibly implemented in clinical

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routine, given the practical constraints of real-world laryngoscopy, including variability in anatomy, motion, and hardware limitations. This serves as a necessary first step toward developing clinically targeted tools, by first establishing a robust and generalizable keypoint tracking framework.

2 | Materials and Methods

Written informed consent was obtained from all participants before study participation. The study was approved by the Institutional Review Board of Mass General Brigham.

2.1 | Dataset

In this study, we developed a CV model to track laryngeal structures using video data. The process involved training the model on one set of videos, where the keypoints of interest were clearly annotated. We then tested the model's ability to recognize these points in a stable way, with a separate set of videos to see how well it performs in new situations.

For training of the tracking model, laryngoscopy videos were utilized retrospectively from multiple sources. The first one was collected from 2019 to 2024 in patients who presented for clinical evaluations to Massachusetts Eye and Ear, using flexible distal chip endoscopes. Tasks included sustained vowel vocalization (/i/), speech, breathing, and sniffing tasks with the glottic and supraglottic structures in view. Patients with various diagnoses were included: 16 with vocal fold immobility (VFI), 5 with muscle tension dysphonia (MTD), 4 with presbyphonia, 3 with laryngeal dystonia (LD) [21], and 19 healthy volunteers. The collected videos were 1920×1080 pixels and 30 fps. The average duration was 56 s. In addition, a total of 8.5 min of demonstrative laryngoscopy videos were extracted from YouTube under fair use to enrich the dataset used for training, with video frames from multiple laryngoscopy cameras and clinic environments. These videos ranged from 320×240 to 1280×720 in size, were 30 fps, and had an average duration of 183 s. A total of 57 videos were extracted (Table 1).

To evaluate the trained model, we assembled an independent test dataset of 10 patients with LD, originally collected as part of the NCT05216770 study conducted at Massachusetts Eye and Ear. The videos were 1920×1080 pixels, 30 fps, and had an average duration of 576 s. This second dataset consisted exclusively of previously unseen LD patients whose outcomes were underrepresented in the training dataset (Table 1).

Key frames representing essential laryngeal motions were extracted and annotated with the DeepLabCut 2.3.6 software [22] by a laryngologist. Thirty-nine anatomically defined keypoints were annotated to represent regions of the larynx (Figure 1A), including the glottis, supraglottis, and midline—expanding upon our prior 13-point glottic map [12]. Keypoints were placed based on spatial relationships, equidistantly along the vocal folds, perpendicularly for false cords, and symmetrically across the aryepiglottic folds and RAE/LAE regions. While not designed specifically for glottal opening analysis, this expanded layout enables more detailed, anatomically meaningful tracking and supports future applications such as assessing supraglottic activity, false cord involvement, and midline symmetry in VFI and LD patients. In our study, a total of 1394 video frames (Figure

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1B) were annotated for the first dataset, and 50 frames for the second dataset. For training of the vocal fold tracking model, the first dataset was randomly separated into 90% training data (1254) and 10% validation data (140), and the second dataset acted as the independent dataset.

2.2 | AI Model Training and Validation

We conducted three experiments: (1) model training and within-study validation, (2) external test set evaluation, and (3) in-clinic feasibility test. The first experiment focused on training the model, where we ran the tracking model over 5,000,000 iterations while creating model checkpoints every 50,000 iterations. The DeepLabCut model was trained using an EfficientNet-B0 [23] architecture, where the backbone was pretrained with the ImageNet database [24]. DeepLabCut implementation was chosen for this study because it allows flexible, user-defined keypoint detection without relying on predefined human landmarks and has been extensively validated in biomedical research. The architecture of EfficientNet is based on convolutional neural networks, which are known to perform well for geometric localization tasks involving multiple keypoints, such as the laryngeal structures. The B0 variant is the smallest in the EfficientNet family, containing only 5.3 million trainable parameters while achieving high accuracies for CV tasks, making it fast enough for our objective of real-time tracking. Input images were resized to 224×224 pixels and normalized using ImageNet mean and standard deviation statistics. The training process utilized the Adam optimizer with an initial learning rate of 0.0005, which followed a cosine decay schedule. A batch size of 8 was used. The model was trained on an Nvidia H100 GPU.

To enhance generalization, mirrored copies of the full training dataset were generated, and a series of data augmentation techniques supported by DeepLabCut were applied during training. Contrast adjustments were performed using CLAHE (contrast limited adaptive histogram equalization) [25] and histogram equalization, each applied with a probability of 0.1. Random rotations up to 180° were introduced with a probability of 0.4, while cropping was applied with a ratio of 0.4.

Scale jittering was implemented with lower and upper bounds of 0.5 and 1.25, respectively. Embossing was enabled with a probability of 0.1. The model incorporated location refinement via Huber loss, with a locref loss weight of 0.05. The total loss used during training was calculated as the sum of the cross-entropy loss between target keypoint maps and predicted keypoint maps and the locref loss. All used training parameters are listed in Table 2.

After model training, the final model weights were obtained, and the prediction performance was assessed by calculating the detection metrics precision and recall, and the localization metric mean keypoint accuracy (mKA) using the 10% validation dataset. The metrics were computed as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

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$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

and

$$\text{mKA} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}$$

For detecting keypoints as true positives with mKA, two parameters were considered during inference. The first was the probability threshold, how confident the model needs to be in order for a prediction to be considered as a keypoint. The second parameter was the Euclidean distance threshold, for which the predicted keypoint needs to be within to be considered as a true positive. A Euclidean distance threshold of 40 pixels was chosen while calculating mKA, as it represented less than 4% of the target image resolution (1080 pixels). Additionally, with an average pixel distance of 206.7 pixels between laryngeal structures and 77.4 pixels between keypoints within a structure, the likelihood of geometric overlap in the analysis was minimal. The prediction performance Table 3 includes group-wise mean values for laryngeal structures and an overall micro-average score across all keypoints.

To assess the temporal consistency of the model's keypoint predictions, we computed the mean Temporal Consistency (mTC) score. This metric quantifies how stable the predicted keypoint positions are over time, independent of true anatomical motion (Figure 1C,D). A lower mTC score indicates smoother, more stable keypoint trajectories, suggesting that the model produces temporally coherent predictions across frames.

To isolate model-induced instability from real patient movement, we constructed a synthetic "consistency video" derived from a representative test frame containing all 39 keypoints. This frame was shifted horizontally at a constant rate of one pixel per frame over 1920 frames. Since the underlying motion is precisely known and uniform, any variation in predicted keypoint positions between adjacent frames reflects tracking jitter or noise introduced by the model. With this video, the *mTC* metric was computed as

$$\text{mTC} = \frac{1}{N} \sum_{b=1}^{B} \sum_{i=1}^{T-1} D_i^b$$

where N is the number of valid frame-to-frame transitions where the detection probability is above the probability threshold, B is the number of tracked points, T is the total number of frames, and D_i^b is the Euclidean distance of body part b between consecutive frames i and $i + 1$, computed only if the detection probability meets the probability threshold. Different combinations of Euclidean distance threshold and probability threshold were investigated in our first experiment to derive the final parameters for the metrics used.

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2.3 | Independent Validation

In the second experiment, we evaluated the model's performance on an independent test dataset by applying the same metrics and hyperparameters defined in the first experiment. This test dataset was deliberately selected as an out-of-distribution example, as it consisted exclusively of patients with LD from a separate study, a group that was nearly absent from the training data, which included only one LD patient. Although the patients in both datasets performed similar vocal tasks during laryngoscopy, the symptomatology in LD is clinically distinct. This evaluation thus serves as a meaningful test of the model's generalizability and robustness to new clinical conditions.

2.4 | In-Clinic Experimentation

In our final experiment, we implemented the proposed model as a Python program with a graphical interface that showcases the real-time video feed, where the keypoint predictions calculated by GPU acceleration were overlayed on top of the video. This program was used with a laptop featuring AMD Ryzen 98945HS CPU, 32 GB RAM memory, and Nvidia RTX 4070 Laptop GPU. Our system was used in a laryngoscopy clinic environment by capturing the 640×480 pixels live video from an Olympus ENF-VH laryngoscope system during a healthy volunteer's examination, producing a video feed of 30 fps. This performance was influenced by three key factors: the number of parameters in the model, image resolution, and the number of floating-point operations (FLOPs) required per forward pass (prediction for one image). The aim of this experiment was to assess the feasibility of real-time analysis by inspecting the frames per second output of the Python program and having a laryngologist conduct a standard examination using the system.

3 | Results

In first experiment, a model checkpoint was selected based on the root mean square error (RMSE) of the validation dataset, and the mTC of the consistency video. Finalized checkpoint 4,500,000 performed the best, resulting in an RMSE of 1.19 per keypoint and mTC of 5.85. Figure 2A shows that the training data learning curve was plateauing at this point, so the following checkpoints performed worse on the validation dataset, giving signs of overfitting. This was also evident in the temporal consistency test (Figure 2B), where the mTC error starts to increase again after this model checkpoint. In this test, when inspecting the X and Y trajectories of each body part (Figure 2C) produced with our finalized checkpoint, left-to-right translation was detected with minimal variability, when the full larynx was shown in the image.

At an optimal probability threshold of 0.5, precision was 85%, recall 93% and mKA 92% for the training dataset; precision was 86%, recall 86% and mKA 85% for the validation dataset (Table 3).

For the second experiment, the model achieved a precision of 97%, recall of 76%, and mKA of 75% for the independent LD test dataset (Table 3). When comparing the results across the three datasets, the training and test datasets exhibit similar precision. However, the LD dataset showed an improved precision of 97%, indicating fewer false positives. On the other

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hand, the recall for the LD dataset was 76%, lower than the 86% recall for the test set and 93% for the training set, highlighting a higher number of false negatives.

In the third experiment, the laptop containing the tracking software was placed in the laryngologist's field of view during a routine examination (Figure 3). The procedure commonly involves viewing the video feed of the laryngoscope, so disruptions to the clinical workflow were minimal. The model successfully ran at 30 fps, matching the captured video framerate. According to the TensorFlow profiler tool, DeepLabCut's implementation of EfficientNet-B0 has 5.3 million parameters and 1.55 billion FLOPs per prediction call. With 30 fps, this amounts to 46.4 billion FLOPs per second, making the model lightweight and well-suited for use on typical laptop hardware. This is consistent with common definitions of lightweight models, which typically have 5 million parameters or less [26–28].

4 | Discussion

Our proposed model demonstrated strong performance tracking laryngeal keypoints in a dataset including normal laryngeal structure as well as patients with VFI, presbyphonia, and MTD. Earlier CV models outside our group are not directly comparable, as most perform segmentation [9–11, 15, 16, 18], classification [7, 8], and the only model with partial keypoint overlap is not publicly available [17]. Comparing to the previous AGATI model, our method performs better in the task of tracking left and right true vocal folds, and the anterior commissure (Supporting Information). For LD patients in the test dataset, the model achieved excellent results in minimizing false positive keypoint detections. The recall of 76% suggests that the model is successfully identifying most relevant keypoints, with further refinements likely enhancing its ability to capture keypoints in underrepresented disorders. Precision reached up to 97% for LD patients, likely due to clearer anatomical landmarks and more consistent motion patterns compared to conditions like VFI. These factors may enhance spatial and temporal feature availability, improving keypoint detection. Future work will analyze motion dynamics and structural artifacts across laryngeal disorders to better understand their impact on model performance. From the tracked laryngeal structures, the cuneiform and corniculate prominences had the most variability. We suspect that, compared to the other structures, supraglottic structures exhibit more variation between individuals.

In terms of computation in a clinic environment, our model is highly lightweight, being capable of running with most laptop-level hardware at 30 fps, matching the input video stream framerate. Future experiments could include capturing a higher-speed video source. For example, doubling the floating-point operations when processing 60 fps video should still be viable for most laptop GPUs. The clinician was able to visually assess the laryngeal structures of the patient while confirming the quality of the tracked points on the screen (Figure 3). Importantly, the clinician reported that the use of the AI model did not interfere with the clinical workflow. The immediate feedback loop gave the clinician more evidence to review in real time, allowing for more comprehensive assessments, which have the potential to result in fewer follow-up visits and faster diagnosis.

Currently, assessments of vocal fold motion and laryngeal function are often subjective, relying on the clinician's interpretation of the video feed. While certainly not designed

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to replace clinician judgment, our AI-assisted model has the ability to enhance video interpretation by providing reproducible keypoints that can be consistently analyzed. Accurate detection of keypoints in real time also proposes the possibility of deriving novel outcome-specific metrics for the identification and quantification of such outcomes. For example, detecting irregular, sudden glottal opening changes captured on video has the possibility of being used to aid the diagnosis of outcomes such as LD [21]. Similarly, outcome-specific metrics can also be developed for lesions, such as polyps, cysts, or even early-stage cancers [29].

Some limitations of this study need to be addressed. While video data from multiple sources were used during training, more data sources should be included in future studies to further improve model generalizability. This includes more data from online sources and different clinics using varying laryngoscopy hardware. While the validation and independent test datasets are of limited size for fully characterizing model performance, they are sufficient to demonstrate the feasibility of applying this AI method in a clinical setting. Future studies should also include more variability in terms of patient outcomes by analyzing more structurally novel outcomes such as vocal cord lesions.

5 | Conclusion

This study proposed a novel keypoint detection model for tracking laryngeal structures in real-time. The model achieved promising performance for the task of tracking laryngeal structures in terms of detection, localization, and temporal consistency. Our model is lightweight and applicable to various hardware, reducing the need to acquire costly computational resources in clinics. In addition to tracking, our model enables future outcome-specific metrics to be developed based on detected keypoints and reported in real-time.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Conflicts of Interest

Kristina Simonyan receives funding from the National Institutes of Health (R01NS088160, R01NS124228, R01DC011805, R01DC012545, and R01DC019353, P50DC01990) and serves on the Scientific Advisory Board of the Tourette Association of America and the Voice Foundation. The remaining authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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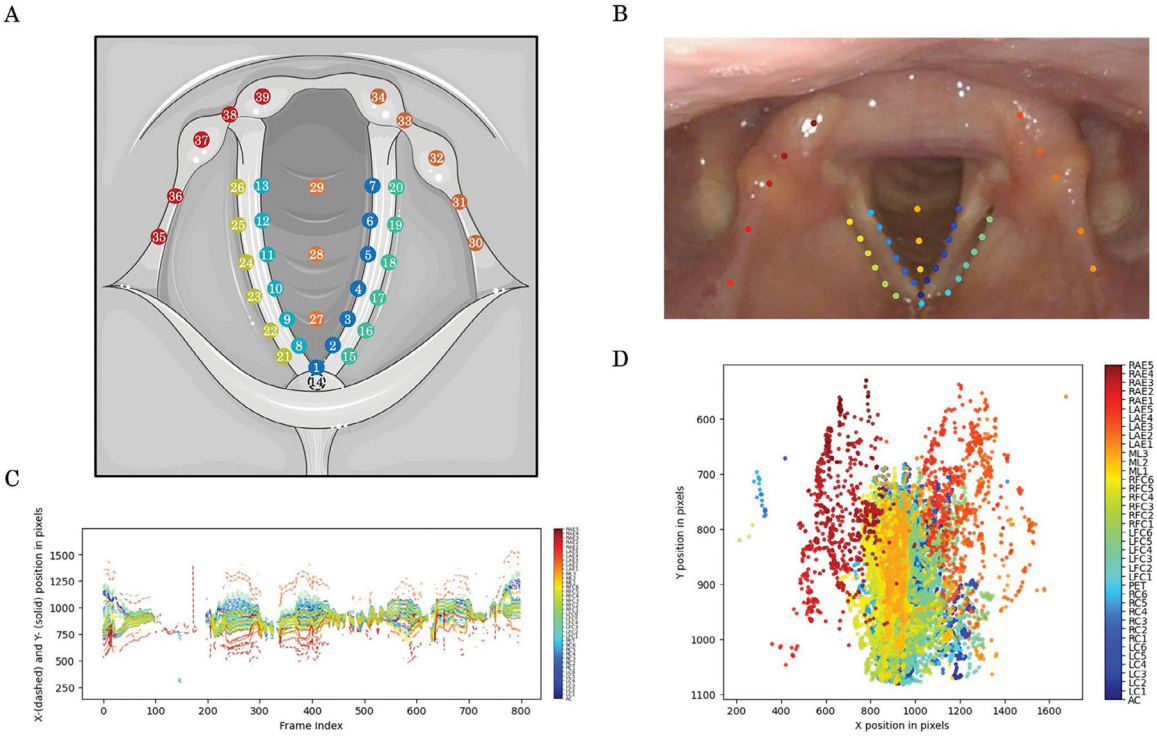

FIGURE 1 |.

(A) Keypoint mapping of the larynx. The left true vocal fold (1–7), right true vocal fold (8–13), midline supraglottis behind the epiglottis (14), false left vocal fold (15–20), false right vocal fold (21–26), midline points (27–29), left aryepiglottic fold (30 and 31), left cuneiform prominence and corniculate prominence with their midpoint (32–34), right aryepiglottic fold (35 and 36), and right cuneiform prominence and corniculate prominence with their midpoint (37–39). (B) An example figure of one frame of a larynx with overlayed keypoint annotations, (C) the body part X and Y trajectories over time, and (D) the X and Y coordinates of body parts in a spatial manner.

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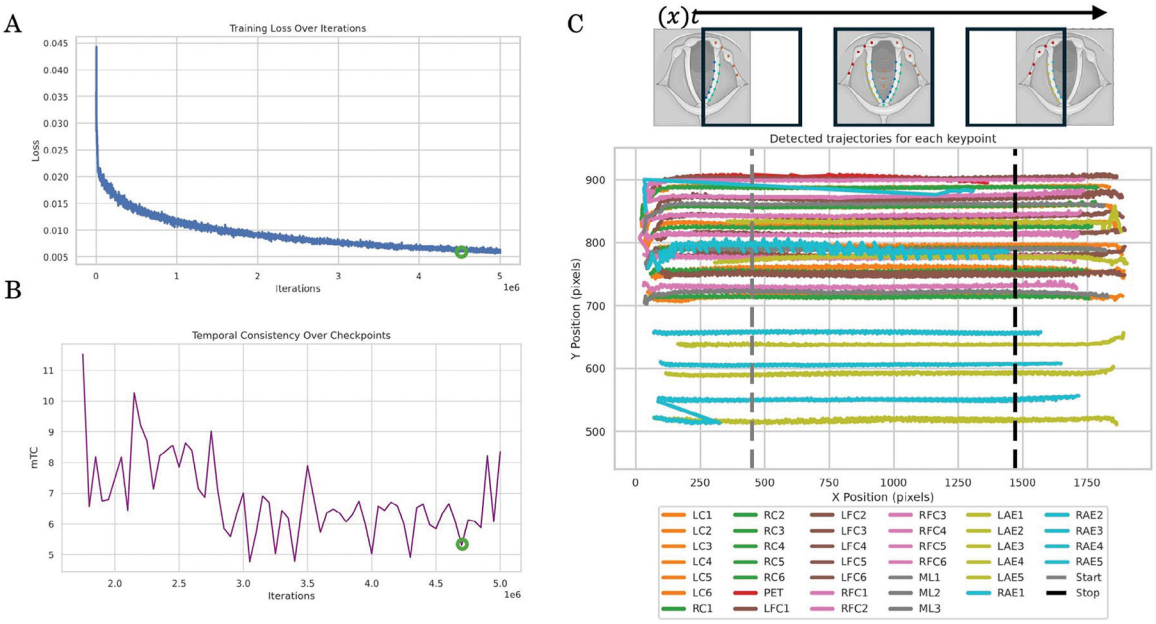

FIGURE 2 |.

(A) Training loss of the model over 5,000,000 iterations where our final checkpoint is highlighted in green, (B) mean temporal consistency of the trained model over saved checkpoints and our final checkpoint highlighted in green, and (C) diagram of the temporal consistency video and the predicted trajectory results of each keypoint during translation. Start and stop are depicted as vertical lines, showcasing the timepoints when the full image of the larynx was on the screen.

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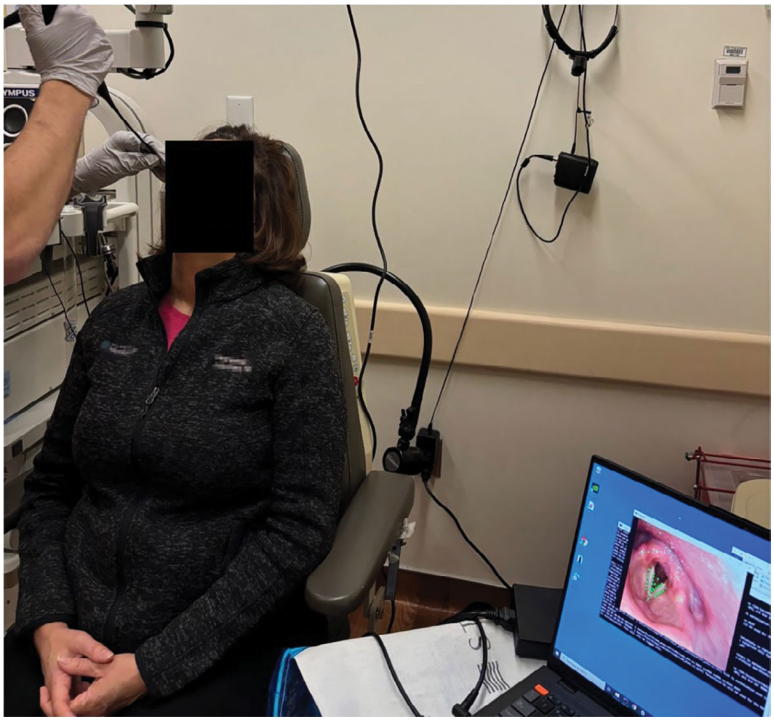

FIGURE 3 |.
A patient undergoing a laryngoscopy while our proposed tracking model is applied in real-time. The video feed from the Olympus ENF-VH laryngoscope system is captured by a laptop and provided as input to the model.

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TABLE 1 |

Demographic and video characteristics of study participants. Data from the 2019–2024 study and publicly available YouTube videos were pooled and randomly partitioned into training (90%) and validation (10%) datasets.

	Training and validation, 90%/10%		Independent test
	Massachusetts Eye and Ear, 2019–2024 study	**YouTube videos**	**Massachusetts Eye and Ear, LD study**
Nobs, videos	47	10	10
Age, mean (1 SD)	61.8 (16.6)	NA	49.3 (8.2)
Gender			
Female	25	NA	9
Male	21	NA	1
Not available	1	NA	0
Outcomes			
Healthy volunteer	19	NA	0
Vocal fold immobility	16	NA	0
Muscle tension dysphonia	5	NA	0
Presbyphonia	4	NA	0
Laryngeal dystonia	3	NA	10
Video duration, mean (s)	56.3	183.1	575.8
Extracted video frames	1192	202	50

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TABLE 2 |

Model training parameters.

Parameter	Value
Max iterations	5,000,000
Save iterations	50,000
Batch size	8
Optimizer	Adam
Learning rate	0.0005
Cosine decay, decay steps	30,000
Cosine decay, α_R	0.02
Cosine decay, decay rate	0.0001
Stride	8
Locref, loss weight	0.05
Locref, SD	7.2801
Global scale	0.8
Rotation augmentation, degrees	180
Rotation augmentation, probability	0.4
Crop augmentation, probability	0.4
Contrast augmentation—CLAHE, probability	0.1
Contrast augmentation—histogram equalization, probability	0.1
Jitter augmentation, bounds	(0.5, 1.25)
Jitter augmentation, probability	0.4
Embossing augmentation, alpha bounds	(0.0, 1.0)
Embossing augmentation, strength bounds	(0.5, 1.5)
Embossing augmentation, probability	0.1

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TABLE 3

I Prediction results of our proposed model with different datasets. Overall and per laryngeal structure metrics are presented, with their bootstrapped 95% confidence intervals.

Laryngeal structures	Training data			Validation data			Test data		
	mKA % [CI]	Recall % [CI]	Precision % [CI]	mKA % [CI]	Recall % [CI]	Precision % [CI]	mKA % [CI]	Recall % [CI]	Precision % [CI]
All (micro average)	92 [91, 93]	93 [92, 94]	85 [83, 87]	85 [81, 88]	86 [83, 89]	86 [81, 90]	75 [71, 78]	76 [72, 78]	97 [96, 98]
Left true vocal fold and AC (1–7)	98 [98, 98]	98 [98, 98]	85 [84, 85]	94 [92, 95]	94 [92, 95]	84 [82, 85]	75 [65, 84]	75 [65, 84]	91 [82, 96]
Right true vocal fold (8–13)	98 [98, 98]	98 [98, 98]	85 [85, 86]	94 [93, 95]	95 [93, 96]	86 [85, 88]	80 [73, 85]	80 [73, 86]	94 [89, 97]
Midline supraglottis (14)	98 [97, 98]	98 [97, 98]	85 [85, 86]	93 [92, 95]	94 [92, 96]	86 [84, 87]	77 [69, 84]	77 [69, 84]	94 [90, 98]
False left vocal fold (15–20)	97 [96, 98]	97 [97, 98]	86 [86, 87]	93 [92, 94]	94 [92, 95]	86 [85, 88]	81 [74, 87]	81 [74, 87]	95 [92, 97]
False right vocal fold (21–26)	97 [96, 97]	97 [97, 98]	86 [86, 87]	93 [92, 94]	94 [93, 94]	86 [85, 87]	82 [76, 86]	82 [77, 86]	96 [93, 97]
Midline points (27–29)	97 [96, 97]	97 [97, 98]	86 [85, 87]	93 [92, 94]	94 [93, 95]	87 [86, 88]	82 [78, 86]	83 [78, 87]	96 [94, 98]
Left aryepiglottic fold (30 and 31)	95 [93, 97]	96 [93, 97]	86 [86, 87]	91 [88, 93]	92 [89, 94]	87 [86, 88]	78 [69, 84]	78 [70, 85]	93 [86, 97]
Left cuneiform prominence and corniculate prominence (32–34)	94 [91, 96]	95 [92, 97]	85 [85, 86]	89 [86, 92]	90 [87, 93]	87 [86, 87]	78 [71, 84]	78 [71, 85]	93 [87, 97]
Right aryepiglottic fold (35 and 36)	93 [90, 96]	93 [90, 96]	86 [85, 86]	87 [82, 91]	88 [84, 92]	86 [85, 87]	74 [65, 82]	74 [65, 81]	90 [83, 96]
Right cuneiform prominence and corniculate prominence (37–39)	92 [89, 94]	93 [90, 95]	85 [85, 86]	86 [81, 90]	87 [83, 91]	86 [85, 87]	73 [65, 81]	73 [65, 81]	91 [84, 96]